Learning conditional language in continuous image space

Imagized Language Model Project

Abstract

Written language includes forms that cannot be faithfully represented by a fixed character inventory. We study language modelling with writing images as both input and output, continuous internal states and one consumer GPU. A causal Chinese raster reader ranks the next glyph at 19.97% top-1 on 2,048 development windows, above unigram, bigram and shuffled-history controls. A normalized autoregressive raster density preserves all 256 supplied development images under greedy copying. Joint reader–density training then improves natural-target likelihood over a matched frozen-reader continuation: 7.026 versus 7.579 nats per image, a paired gain of 0.553 with 95% source-record interval [0.450, 0.665]. Ordered earlier history supplies a further 0.185-nat advantage over shuffling. The 18.71M-parameter joint runtime completes 8,000 additional updates in 342.4 seconds, peaking at 0.648 GiB allocated CUDA memory, excluding foundation training. Autonomous target identity remains only 9/256, versus 8/256 with earlier history shuffled, and the registered generation criteria fail. These results demonstrate reliable visual copying and learned conditional image likelihood on one GPU, while leaving coherent language generation and broad instruction following unresolved.

Written language is encountered as marks on paper, screens and historical surfaces. Its visual form carries information about script, handwriting and material history. Modern character encodings provide a powerful interface to many such marks, but a finite inventory cannot express every damaged, transitional or unencoded form. An image-based language model could retain that evidence throughout reading and generation, with transcription offered as an optional output view.

Pixel-based language modelling already has substantial precedents. PIXAR demonstrates autoregressive language generation in pixel space and identifies the difficulty of generating readable text[1]. Continuous autoregression can use a diffusion loss in place of vector quantization[2], while Diffusion Forcing models continuous sequences under independently chosen noise levels[3]. These results motivate a visual language interface, but do not establish that a small glyph generator acquires linguistic knowledge or that image input automatically reduces training cost. Exact pixel factorization also has established precedents in PixelRNN and conditional PixelCNN[6,7]. Masked autoregressive networks provide normalized probabilities without a character dictionary[8]. Multilingual generation from pixels has also been studied[9].

We investigate the conditional generation problem directly. The model receives raster history, predicts a distribution of future writing, emits pixels and rereads those pixels. Chinese provides a demanding initial case: stroke configurations must remain coherent while context selects among many similar written forms. The present study uses one canonical font to isolate this coupling before introducing handwriting or historical scripts.

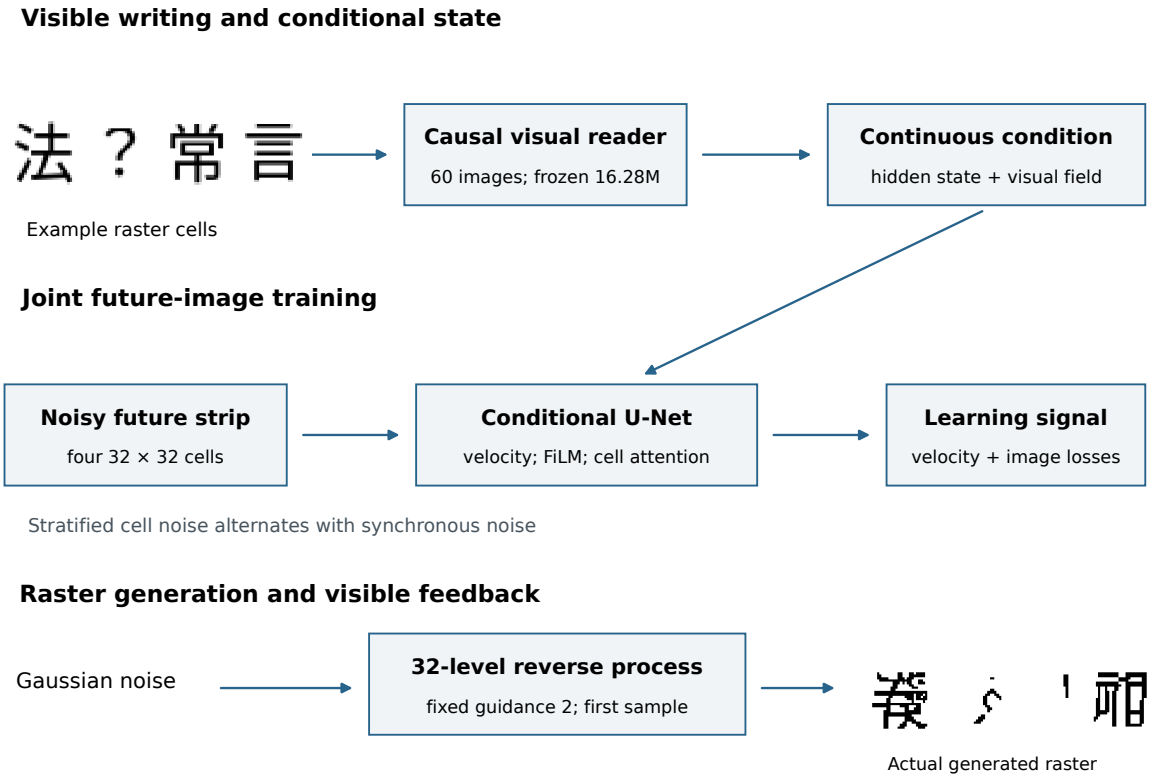


Fig. 1 | A visual language interface and its conditional generation test. a, Rendered or native writing supplies ordered image cells to a causal visual reader. Text rendering is an optional host interface. **b**, A conditional denoiser uses the frozen reader state and a noisy four-cell strip to predict diffusion velocity. Future cells exchange information through convolutions and pooled-cell attention. **c**, Sampling starts from noise, emits four visible cells and appends their pixels to the next context. Optional transcription follows generation and is absent from the model. The schematic shows the implemented experimental contract, without claiming that its generated content is already correct.

Results

A continuous visual interface preserves the written signal

Each observed cell is a 32×32 ink image. An orthonormal transform maps its signed pixels into a 1,024-dimensional continuous field. The transform is invertible and contains no learned character dictionary. An ordered sequence of these fields enters a causal transformer. The deployed model receives no Unicode values, OCR transcripts, character IDs or vocabulary logits (Fig. 1). Host-side rendering uses encoded source text during data preparation. This distinction matters: the current benchmark studies learning from rendered Chinese, rather than discovering a script from raw archaeological photographs.

Ordered image history contains next-writing information

The strongest established reader result uses 2,048 deterministic development windows and an evaluator-only bank of 1,024 canonical glyph images. Full-history next-image top-1 is 19.97%, above the image unigram at 1.42%, the symbolic bigram at 12.26% and the prefix-shuffled condition at 18.36%. The ordered history improves target log probability by 0.174 nat over the shuffled control. This result supports bounded conditional prediction from image sequences.

These figures measure selection among completed images. The evaluator bank does not enter the model and cannot act as a deployed writer. Thresholding the predicted continuous field can mix strokes from several plausible futures. For squared-error prediction, the population optimum is $f^*(H) = \mathbb{E}[X \mid H]$. A conditional mean need not be a valid glyph when several futures are possible. Sharp copying, target ranking and sampled language generation therefore require separate measurements.

A glyph prior does not automatically bind to language

To separate visual synthesis from contextual prediction, we train an unconditional glyph prior and a noisy visual compatibility critic while freezing the causal reader. The combined runtime contains 23.27 million parameters. The critic scores an arbitrary noisy image against the visible history. In the ideal density-ratio limit,

$$\nabla_x \log p_t(x \mid H) = \nabla_x \log p_t(x) + \nabla_x \log \frac{p_t(x \mid H)}{p_t(x)}. \tag{1}$$

The construction is related to classifier-guided diffusion[4].

On 512 development windows, the clean critic reaches 21.875% top-1, versus 20.898% with shuffled history. Its first guided sample reaches zero target identities in 256 cases. Selecting the highest clean critic score among four samples reaches 1/256, with the same result after shuffling history. The generated glyph-like marks are diverse and nonblank, yet fail conditional identity (Fig. 2).

This experiment identifies a reader–writer gap without proving a unique cause. At the terminal corruption level, target signal power is 2.43×10^{-7}, and an exact conditional guidance term should vanish. That limit also occurs in successful conditional diffusion. It cannot alone explain the failure. Insufficient conditional-score accuracy, off-manifold gradients, limited prior quality and finite sampling remain compatible explanations. We consequently test a directly conditional writer rather than interpret clean-image ranking as evidence of a generative solution.

Direct conditional block generation tests the missing coupling

The new writer models a joint four-cell future conditioned on 60 visible cells. It reuses the 16.28M-parameter reader and fine-tunes a 5.90M-parameter strip denoiser. A projected reader state modulates each residual scale, and a four-cell attention layer exchanges future information at the bottleneck. There is no separate guidance critic or candidate selector.

For signed clean writing x_0, noise ε and cumulative signal $\bar{\alpha}_t$, the training target is velocity:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\,\varepsilon, \tag{2}$$

$$v^* = \sqrt{\bar{\alpha}_t}\,\varepsilon - \sqrt{1 - \bar{\alpha}_t} x_0. \tag{3}$$

The clean estimate is $\hat{x}_0 = \sqrt{\bar{\alpha}_t} x_t - \sqrt{1 - \bar{\alpha}_t}\,\hat{v}$. Training alternates independently stratified future noise levels with synchronous levels that cover the sampling distribution. Fixed visual-field

and foreground losses retain the writing signal. Intermediate target-field alignment follows the motivation of representation alignment[5], using a fixed raster transform rather than a semantic teacher.

The run completes all 8,000 updates in 361.6 s, using 0.543 GiB peak allocated CUDA memory. Evaluation uses one primary sample from 256 prefixes under paired initial noise. Ordered first-position identity is 1.17%, compared with 1.17% after shuffling and 0.00% with a zero condition. Across all four future cells, identity is 0.586%, with an ordered-minus-shuffled difference of 0.098 percentage points and a record-bootstrap 95% interval of [0.000, 0.307]. Pixel F1 is 0.320. The registered complete-mechanism decision is `reject`. The raw image panel shows the first fixed examples without reranking or selection (Fig. 2).

A post-result motor diagnostic supplies the intended first-cell image to the same trained writer on 64 fixed prefixes. Exact plan and image-field supply raise pixel F1 from 0.378 to 0.459 and target identity from 1/64 to 5/64. The fixed inverse transform reproduces the supplied raster with F1 1.0. Because this test supplies the answer, it provides no autonomous-language evidence. It shows that the current learned writer still loses substantial shape information even with an exact visual condition.

Exact raster likelihood preserves copying but not conditional language

The diffusion writer's supplied-image failure motivates a distribution that retains the exact inverse visual field as aligned pixel logits. For binary image X, ordered history H and row-major pixel index p, we define

$$p_\theta(X \mid H) = \prod_{p=1}^{1024} \mathrm{Bern}[X_p; \sigma(6A_p + \delta_\theta(H, X_{<p}))]\,. \tag{4}$$

Here A is the inverse predicted field, scaled to signed pixels and clipped to $[-2,2]$. A causal row transformer and within-row MADE supply the residual. The density contains 2,434,592 learned parameters, combined with the frozen reader for 18,711,841 runtime parameters. This is a conditional raw-pixel factorization[6–8], not a new language vocabulary.

With a supplied binary image, A_p is exactly ± 1. A zero-initialized residual therefore copies its greedy mode exactly before training. This is an architectural property, not acquired linguistic knowledge. Pure image NLL training retains that property: greedy copying is exact for all 256 development targets, stochastic-copy F1 is 0.99949, and a deranged supplied image reduces F1 to 0.358 (Fig. 3). These exercises supply the answer and cannot qualify autonomous generation.

On 512 natural targets, ordered NLL is 15.219 nats per image (0.02144 bits per pixel), versus 15.254 with earlier history shuffled. The paired advantage is 0.0355 nat, with source-record bootstrap 95% interval [-0.1014, 0.1693]. This interval includes zero. An independent-pixel distribution using the same predicted plan has NLL 846.09 nats. That comparison supports learned dependencies within the image; it does not isolate learned linguistic dependencies between images.

Autonomous sampling recovers 4/256 nearest target identities, versus 3/256 after shuffling, with

pixel F1 0.350. The paired identity advantage is 0.391 percentage points, with interval [0, 1.210]. Greedy generation reaches 14/256 exact images but only 0.215 F1 and 0.330 target-relative ink, consistent with frequent sparse forms in the visible gallery. Four-step feedback identity is 4.297% ordered and 3.125% shuffled; this small secondary difference does not reverse the failed primary language tests. The registered decision is `reject`.

Joint learning improves the conditional image distribution

A matched experiment tests whether the same raster law benefits from learning its reader under image NLL. Both arms start from the exact-density EMA and receive identical new training draws. One continues to freeze the reader; the other differentiates likelihood through its hidden state and predicted field. Reader dropout remains disabled in both, so gradient flow is the controlled change. The unused contrastive temperature stays fixed.

After 8,000 updates per arm, joint target NLL is 7.026 versus 7.579 for the frozen control. The paired gain is 0.553 nat, with interval [0.450, 0.665]. Joint ordered history improves NLL over earlier-shuffled history by 0.185 nat, with interval [0.128, 0.244]. The frozen continuation also gains an earlier-history effect (0.119 nat, interval [0.052, 0.186]); extra training alone therefore explains part of the progress. The matched comparison isolates the additional likelihood benefit of updating the reader.

Joint first-sample identity is 9/256, versus 6/256 for the matched control and 8/256 after shuffling. The joint-minus-control identity difference is 1.172 percentage points, with interval [-0.397, 2.893]. Both arms retain exact greedy copying of all 256 supplied images. Joint sampling has F1 0.362 and isolated-ink fraction 0.00744. The joint arm passes six of eight gates, but fails sampled quality and ordered-generation criteria; its decision remains `reject` (Supplementary Note 13 and Fig. S2). This is evidence of improved conditional learning, not a qualified language generator.

Discussion

The results separate a useful visual predictive state from a useful visual language generator. The established causal reader learns context-dependent next-writing information, while subsequent writers can erase that information or model the marginal appearance of glyphs. A complete visual language model must bind meaning and sequence to the exact marks it generates.

An image interface enlarges the representable output space, including the possibility of forms absent from a character encoding. It does not remove the need to learn syntax, knowledge or long-range dependencies. This study has not demonstrated general instruction following, etymology answers, handwriting transfer or arbitrary historical-script generation. It also lacks independent training-seed replication and a capability-matched text-model comparison. Small allocated memory in a bounded experiment establishes feasibility on one GPU, rather than a general efficiency advantage over language models.

Exact copying is now preserved by an aligned inverse-image path. The remaining test is whether context selects a coherent written future. Low teacher-forced image NLL can reflect dependencies

among strokes already revealed within the target, while generated output still loses global identity. Joint training improves proper likelihood beyond matched extra density training; the raw generated continuations still fail the intended language test. Conditional continuation and short answers require counterfactual prompts and the same context effect during autonomous sampling and visible feedback. Broader books and offline teacher-generated exercises can supply later training data while leaving the final inference path independent. A larger corpus or longer training requires improvement on the intended generation task, beyond the conditional-likelihood gain established here.

Methods

Data, rendering and partitioning

The public Chinese stream contains 7,017 provenance-tracked records. Records are assigned deterministically to training, development or a sealed partition before rendering. Original and simplified views inherit their source split. Both generation experiments use one fixed Noto Sans CJK font, 26-pixel writing inside 32-pixel cells, without image augmentation. A 65-cell training segment supplies the first 60 cells as context and cells 61–64 as future writing. The remaining cell is unused. No string, source identifier or symbolic target enters a model method. Sealed records remain unopened.

Training and numerical verification

In the diffusion experiment, the reader is frozen. Copied glyph-prior weights initialize the denoiser, which is then optimized as a velocity predictor. AdamW uses a learning rate of 5×10^{-5} for transferred layers and 2×10^{-4} for new modules, 400 warmup updates, cosine decay to 10%, batch size eight, BF16 arithmetic and gradient clipping at one. EMA decay is 0.999. Checkpoints retain model, optimizer, RNG, dataset offset and provenance hashes. Interrupted and continuous smoke runs produce byte-identical model and EMA weights.

The field objective applies an orthonormal transform directly to floating signed predictions. Thresholding is confined to output and identity evaluation. This distinction is verified by an isolated pixel-gradient test. The primary velocity MSE is supplemented by foreground, Dice, field and representation losses at weight 0.10 each, plus an ordered-history margin at weight 0.25. Supplementary Information gives the complete objective and gates.

The exact-density experiment uses 64-cell causal histories and 16 stratified positions in each of eight streams per update. A quarter of selected images receive supplied-image motor conditions; the remainder use predicted states. Binary observations are thresholded once before unweighted Bernoulli NLL. AdamW uses learning rate 2×10^{-4}, 400-update warmup, cosine decay, BF16, norm clip one and EMA 0.999. All 8,000 updates complete in 207.8 s with 0.400 GiB peak allocated CUDA memory. This timing excludes the frozen reader's earlier training. Exact resumed weights and EMA match uninterrupted smoke training. Strict-causality gradient tests and enumeration of a four-bit subproblem verify the probability masks.

The matched continuation reinitializes optimizer and EMA from identical exact-density weights.

Both arms make 8,000 additional updates, using the same data and training generator. The joint arm uses reader learning rate 2×10^{-5} and density rate 2×10^{-4}, with the same schedule and motor mixture. It trains 18,711,840 participating parameters; the frozen arm trains 2,434,592. Joint and control times are 342.4 and 218.0 s, with peaks 0.648 and 0.400 GiB. Frozen-reader weights remain identical to initialization, while joint reader weights change. Both final training RNG states match. Development opens only after both arms finish.

Sampling, controls and statistics

The diffusion experiment uses 32 registered DDIM schedule levels and classifier-free scale two. Four future cells start from Gaussian noise. The same noise is reused for ordered, suffix-preserving shuffled and zero-condition histories. Only one sample contributes to the primary block result. Generated cells are appended as visible rasters for a second four-cell block.

An evaluator-only canonical bank measures nearest-raster identity after generation. Targets outside the bank receive no identity credit, and bank coverage is reported per position. This measurement can assign a nearest identity to an illegible pseudo-glyph, so direct images and surface statistics accompany every result. Pixel F1 alone is also insufficient because many Chinese glyphs share strokes. No OCR is used by the model or the reported identity evaluator.

The exact-density audit uses 512 NLL targets, 256 single-image generations and 64 four-step feedback streams. Its sampler uses 32 row-transformer evaluations and 1,024 masked-head evaluations per image. Ordered and shuffled runs share the same Bernoulli uniforms. Its post-generation bank includes all distinct training forms plus every audit target, including punctuation: 9,932 unique binary rasters and 100% target coverage. This enlarged bank differs from the earlier Han-only bank. Zero condition is an out-of-distribution intervention, not a separately trained unconditional baseline. One standalone export embeds the complete reader and density; image-prompt inference reproduces the training-checkpoint continuation exactly under the same seed and greedy mode.

Confidence intervals use 2,000 bootstrap resamples of source records, retaining all windows from a record together. Paired differences use identical windows and initial noise. Intervals describe development-record variability for this trained checkpoint, not variability across training seeds. The current development partition has informed a sequence of experiments and is not a final independent generalization test.

Data and code availability

Implementation, protocols, evidence summaries and scripts are available in `https://github.com/lachlanchen/ImagizedLanguageModel`. Manifest and checkpoint hashes identify the exact local research artifacts. Restricted reference books are not redistributed. The complete research record and Supplementary Information distinguish earlier foundation provenance from the present raster-only deployment experiment.

References

1. Tai, Y., Liao, X., Suglia, A. & Vergari, A. PIXAR: Auto-Regressive Language Modeling in Pixel Space. *arXiv* 2401.03321 (2024). https://arxiv.org/abs/2401.03321.
2. Li, T., Tian, Y., Li, H., Deng, M. & He, K. Autoregressive Image Generation without Vector Quantization. *Advances in Neural Information Processing Systems* (2024). https://arxiv.org/abs/2406.11838.
3. Chen, B. et al. Diffusion Forcing: Next-token Prediction Meets Full-Sequence Diffusion. *arXiv* 2407.01392 (2024). https://arxiv.org/abs/2407.01392.
4. Dhariwal, P. & Nichol, A. Diffusion Models Beat GANs on Image Synthesis. *Advances in Neural Information Processing Systems* (2021). https://arxiv.org/abs/2105.05233.
5. Yu, S. et al. Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think. *International Conference on Learning Representations* (2025). https://arxiv.org/abs/2410.06940.
6. van den Oord, A., Kalchbrenner, N. & Kavukcuoglu, K. Pixel Recurrent Neural Networks. *Proceedings of the 33rd International Conference on Machine Learning* **48**, 1747–1756 (2016). https://proceedings.mlr.press/v48/oord16.html.
7. van den Oord, A. et al. Conditional Image Generation with PixelCNN Decoders. *arXiv* 1606.05328 (2016). https://arxiv.org/abs/1606.05328.
8. Germain, M., Gregor, K., Murray, I. & Larochelle, H. MADE: Masked Autoencoder for Distribution Estimation. *Proceedings of the 32nd International Conference on Machine Learning* **37**, 881–889 (2015). https://arxiv.org/abs/1502.03509.
9. Hu, C., Tai, Y., Vergari, A., Keller, F. & Suglia, A. MIXAR: Scaling Autoregressive Pixel-based Language Models to Multiple Languages and Scripts. *arXiv* 2604.11575v2 (2026). https://arxiv.org/abs/2604.11575v2.

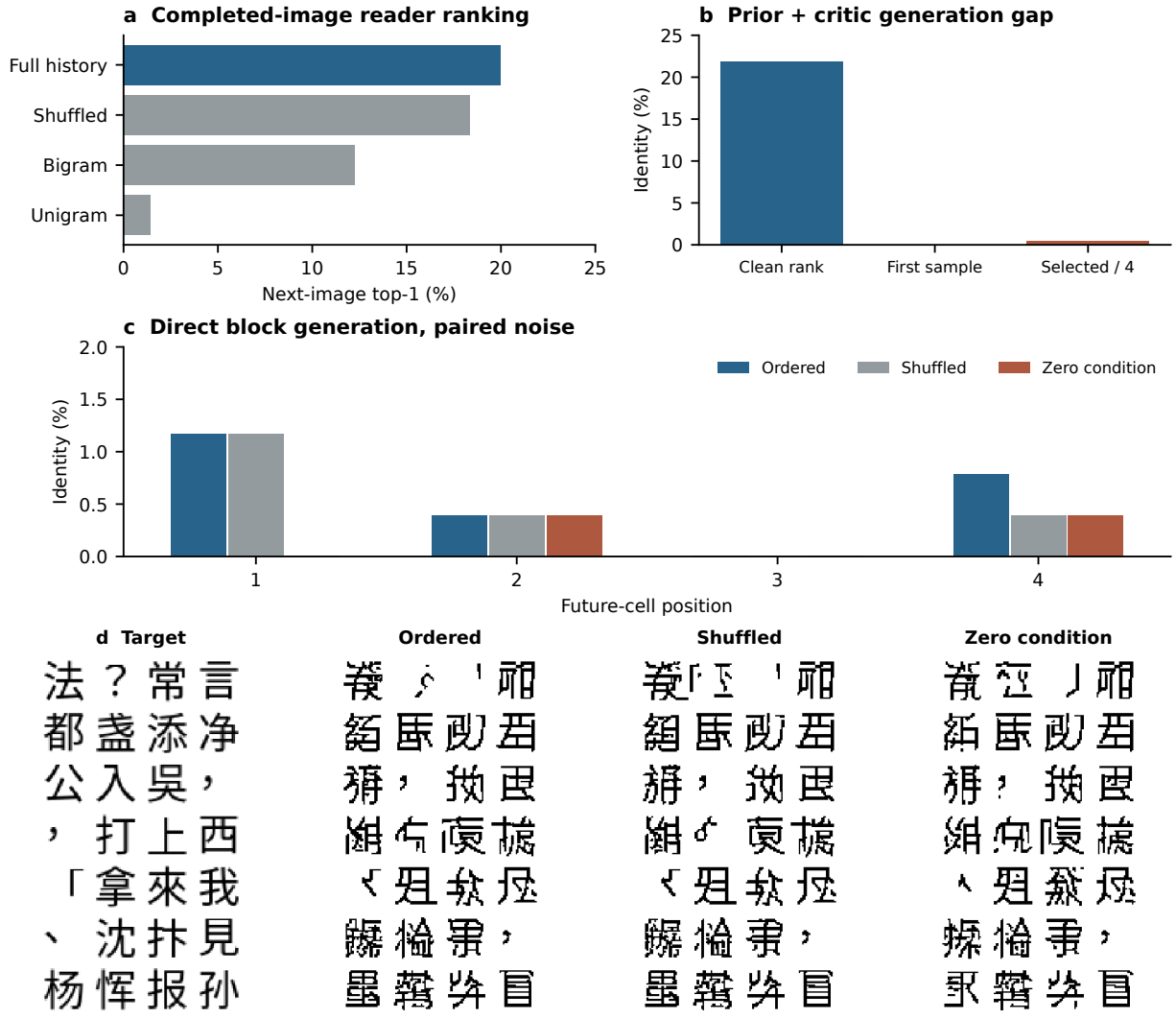


Fig. 2 | Prediction and generated-image evidence. a, The established causal reader ranks completed next images above unigram, bigram and shuffled-history controls on 2,048 windows. These are ranking measurements, not generated text. **b**, Prior-plus-critic guidance retains completed-image ranking but almost never generates the target identity. **c**, Direct conditional block generation reports all four positions under ordered, shuffled and zero conditions. Bars use the same starting noise and single primary samples. **d**, Fixed, unselected examples show the target and all generated controls. Images are raw model outputs with nearest-neighbour enlargement. No OCR or image-bank replacement is applied.

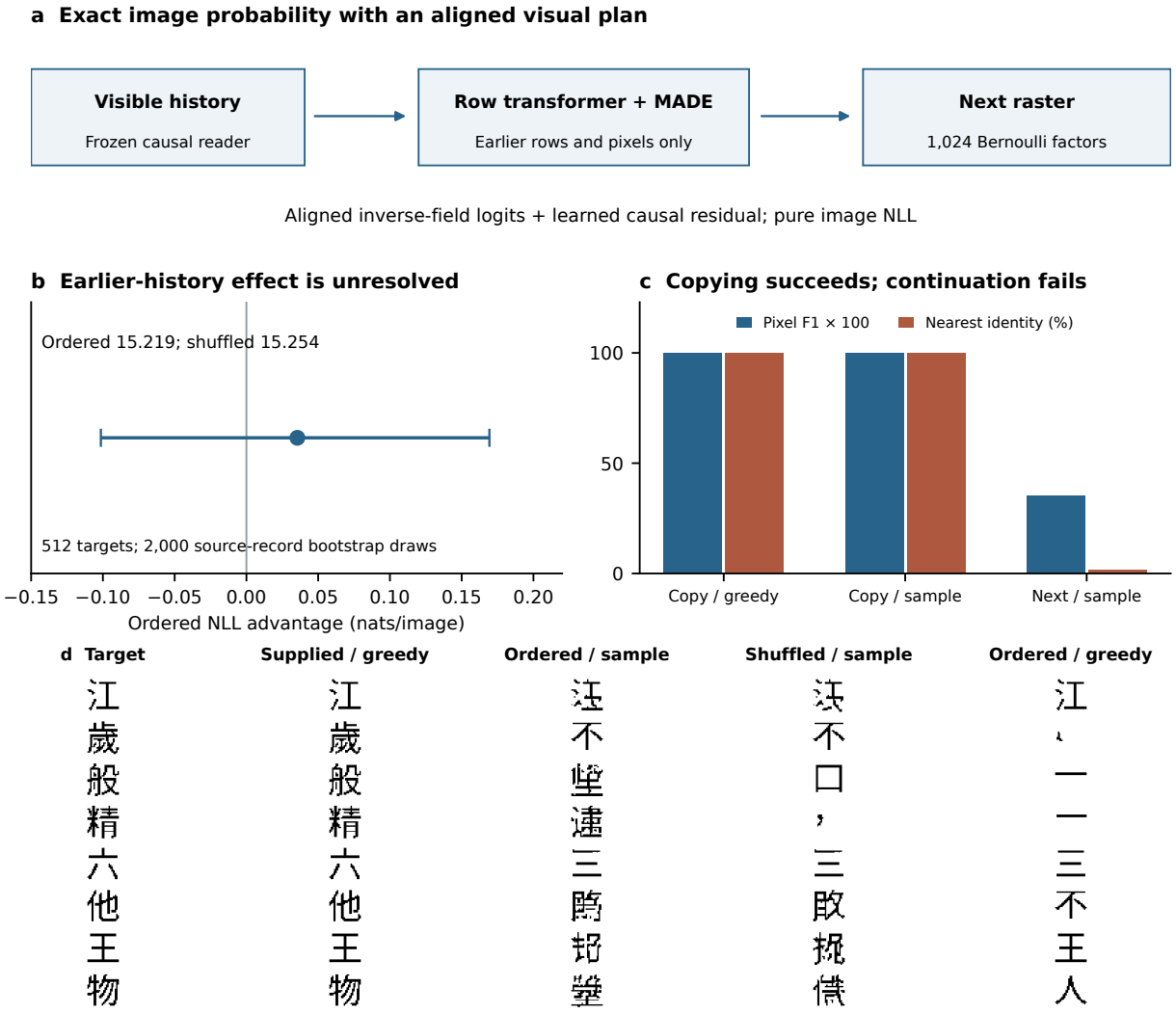


Fig. 3 | Exact image likelihood separates motor success from language failure. a, A frozen image reader supplies an aligned plan and continuous history state. Strict row and pixel masks give a normalized raster law. **b**, Ordered-versus-shuffled NLL advantage on 512 development targets; the interval uses 2,000 source-record bootstrap draws and crosses zero. **c**, Supplied-image copying and autonomous next-image sampling on 256 targets. The copy controls reveal the answer. **d**, The first eight audit rows, shown without selection, compare target, exact supplied-image copying and autonomous controls. Every displayed output is the actual model raster. The 9,932-image evaluator bank is created after generation, covers all targets and never selects or replaces an output. Its identity rates are not directly comparable to earlier 1,024-image-bank results.